```
class Rectangle: public Shape
{
public:
int getArea()
{
return (width * height);
}
};
int main()
{
Rectangle Rect;
Rect.setWidth(5);
Rect.setHeight(7);
// Print the area of theobject.
cout<< "Total area: "<<Rect.getArea() <<endl;
return 0;
}
```

Output
Total area: 35

The following table contain the members defined inside each class before inheritance

MEMBERS of class	visibility modes		
	Private	Protected	Public
Shape(base class)	int count;	int width; int height;	void setWidth(int) void setHeight(int)
Rectangle (derived class only with its defined members)			int getArea();

The following table contains the details of members defined after inheritance

MEMBERS of class	visibility modes –public for acquiring the properties of the base class		
	Private	Protected	Public
Shape(base class)	int count;	int width; int height;	void setWidth(int) void setHeight(int)
Rectangle (derived class acquired the properties of base class with public visibility)	Private members of base classes are not directly accessible by the derived class	int width; int height;	int getArea(); void setWidth(int) void setHeight(int)

Suppose the class **rectangle is derived with protected visibility** then the properties of class rectangle will change as follows

Chapter 16 Page 260-274.indd 267 3/24/2020 9:23:56 AM